Tanmay Bhuskute

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SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) with product engineering experience shipping full-stack web apps and data-heavy systems in Python, TypeScript, React, and Node.js. Prior SWE internships plus current research building and evaluating agentic LLM pipelines.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Data Analytics, Machine Learning, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Python, TypeScript, JavaScript, Java, C/C++, SQL, Go.

Web & Backend: React, Node.js, Express, FastAPI, Django, Flask, REST APIs, GraphQL, HTML/CSS.

Databases: MongoDB, PostgreSQL, MySQL, Redis, DynamoDB.

Developer Tools: Git, GitLab CI/CD, Jenkins, AWS, Docker, Kubernetes, Terraform, Linux, Bash, PySpark, SonarQube, GCP.

AI & Data: LLM evaluation, agentic systems, RAG, MCP, Gemini, GPT-4o, Vertex AI, LangChain, PyTorch, pandas.

WORK EXPERIENCE

Graduate Researcher | University of Cincinnati Feb 2026 – Present

- Designed two-stage memory-augmented LLM pipeline decoupling video understanding from device automation for Android GUI bug reproduction, evaluated against **ViBR** (GPT-4o + GroundingDINO + CLIP baseline) across 53 runs on 20+ apps.
- Built provider-agnostic LLM abstraction (Gemini, GPT-4o, Llama, Qwen) and config-driven eval harness; demonstrated **25.7x latency reduction** while preserving task-completion fidelity.

Software Development Engineer | Siemens Industry Software Jul 2022 – Jun 2024

- Shipped Python (Boto3) automation replacing legacy Ruby workflows for **Teamcenter X**, cutting deployment time by **40%** across AWS environments.
- Executed **44 production environment upgrades** in 4 months, diagnosing and resolving live issues across Linux, EC2, databases, and services with zero interruption; built self-service developer platform adopted by **15 engineers**.
- Migrated **40+ production environments** to Kubernetes using ArgoCD, Helm, GitLab CI/CD, and HashiCorp Vault; designed Blue-Green framework reducing upgrade downtime **50%**.
- Served as Quality Coach and SonarQube Specialist, raising automated code coverage from **35% to 78%**; mentored **5 software engineers** through structured onboarding.

Software Developer Intern | Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**.
- Developed **Go + Terratest** framework, catching **95%** of misconfigurations pre-production.

Software Developer Intern | Government of Maharashtra, Water Resource Department Sep 2021 – Dec 2021

- Built and shipped a full-stack **React and MySQL** workflow application that digitized departmental record management, automated manual processes, and improved operational efficiency by **40%**.

PROJECTS AND RESEARCH

PolicyPilot – Policy Intelligence Portal [React, TS, Node, Express, Gemini, MCP] May 2026

- Shipped full-stack TypeScript app (React SPA + Express REST API + MCP server) with evidence-grounded **RAG pipeline** forcing citation-backed LLM answers for high-stakes eligibility decisions.
- Designed **17-tool MCP server** exposing structured LLM tool-calling consumable by external AI agents, with Zod validation at every data boundary and Vitest unit and integration tests.

NYC Mobility Forecasting [Python, PySpark, Delta Lake, scikit-learn, FastAPI, React] Jul 2026

- Built ML system forecasting demand across **263 zones** over a **2.3M-row** hourly dataset via a PySpark medallion pipeline; two-stage Poisson hurdle model improved MAPE **18.77% to 16.52%** and cut RMSE **43%**, served from FastAPI to a React dashboard with Redis caching.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).